

Genga K

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Applied AI engineer with expertise in building scalable production AI systems - RAG pipelines, agentic workflows with tool calling, evaluation harnesses, and real-time voice and vision services. Owned end-to-end architecture across retrieval design, async backends, guardrails, and observability.

PROFESSIONAL EXPERIENCE

JOBTWINE, AI Engineer

Jul 2025 – Present, Bengaluru, India.

- Architected a multi-signal **AI proctoring** engine that independently scores content, AI-likeness, audio, and video streams and fuses them via weighted blending with exponential boost functions into a calibrated 0–100 suspicion score with risk bands and flagged episodes.
- Spearheaded end-to-end LLM-driven interview tooling: a JD-to-playbook generator (skills + questions), a semantic resume shortlister, and post-interview analytics dashboards with LLM-generated debriefs.
- Built a real-time **RAG** pipeline over ~10–50K chunks of role rubrics and JDs, grounding the interviewer LLM to eliminate context-absence failures.
- Designed **async** FastAPI backend with a 10-worker **SQS** queue and **multiprocessing**, supporting 10–50 concurrent sessions and 1000+ queued requests.
- Shipped a WebRTC voice agent with streaming and turn-based modes, interruption handling, barge-in detection, and multi-turn conversational memory.
- Created a cv pipeline using MediaPipe for gaze tracking, face-presence detection, and lip-sync correlation at ~10 FPS feeding the video scoring module.
- Technologies: **Python, RoBERTa, OpenCV, Dlib, NLTK, Wav2lip, MediaPipe, ASR, TTS, LangGraph, SQS, RAG, Pinecone, Langchain, Multiprocessing, Flask, MCP, Agentic AI, Multi-agents, LLM as judge.**

SPACE MARVEL AI, AI Consultant (Freelance)

Jan 2025 – Jul 2025, Remote, India.

- Led a three-person team across three AI systems; owned architecture and technical direction, set the agent orchestration pattern and ran design reviews.
- Delivered a shelf-auditing system using CV segmentation to localize product clusters and a VLM stage to identify SKU names and counts, driving automated restock alerts across product categories.
- Produced an outbound tele caller bot on an ASR → LLM → TTS pipeline, handling interruptions and classifying call outcomes at 200 calls/day.
- Created a voice-driven desktop agent that reads open browser and window state to automate multi-step daily workflows via tool calling.
- Technologies: **Python, LLM agents, VLM, OpenCV, ASR/TTS, tool calling, FastAPI**

YUBI, Data Scientist Intern

Dec 2024 - Jun 2025, Chennai, India.

- Shipped **OCR + LLM**-based document classification APIs into loan-application processing, automating text and key-value extraction from KYC and financial documents at ~5,000 docs/day with 50,000+ batch runs.
- Developed a **hybrid OCR engine** (rule-based + LLM) that preserved document layout by replicating **pixel coordinates and spacing**, outperforming PaddleOCR, Tesseract, and AWS Textract on complex multi-column PDFs across 30+ document types.
- Engineered validation modules to detect file tampering, verify signature presence and originality, flagging low-trust documents before loan processing.
- Devised an ASR/TTS audio bot to handle customer calls and extract behavioural signals around loan-repayment intent.
- Technologies: **Python, LLM (gpt4o, Llama.cpp), Prompt Engineering, Fine-Tuning, OCR, FastAPI, Multi agent systems, AWS Sagemaker, OpenCV, Transformers, AWS Transcribe, Azure Cognitive Services, OpenAI Whisper, Parallel processing.**

EXL HEALTH, Data Engineer Intern (R&D)

Jun 2023 - Nov 2023, Chennai, India.

- Programmed **Robotic Process Automation** (RPA) applications to **automate** graphical and web-based user interfaces, to eliminate manual intervention
- Conducted **data wrangling** on large volumes of scraped information by applying text mining and **Natural Language Processing (NLP)** techniques to streamline workflows which resulted in **95% reduction in task completion** time.
- Devised a custom **ETL** tool to extract, clean, and load web data, achieving a 60% reduction in processing time.
- Technologies: **Python, Lackey, PyautoGUI, Tesseract OCR, Selenium, HiveQL, RPA, NLP, Data Mining.**

ELAMIGO, Data science Intern

Dec 2022 – Feb 2023, Bengaluru, India.

- Scraped and structured data across ~40 industry sectors into a queryable database, ran EDA surfacing 15+ partnership insights.
- Technologies: **MySQL, Beautiful Soup, Selenium, PowerBI, web hosting.**

KEY SKILLS

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| <ul style="list-style-type: none">Programming Languages: Python, R, Java, C++Web Development: React, JavaScript, HTML, CSSFrameworks & Tools: Streamlit, Flask, FastAPI, DjangoCore Competencies: Data Structures & Algorithms, OS, DBMS | <ul style="list-style-type: none">Automation & Scripting: Selenium, PyAutoGUI, Lackey, VBADatabases: MySQL, MongoDB, Oracle, PineconeData Science & Machine Learning: Predictive Analytics, NLP, Data Mining, Statistical Modeling , Time Series Forecasting, Deep learning.Artificial Intelligence: GenAI, RAG, CI, Agentic AI, Agentic systems. |
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ACADEMIC PROJECTS

Disease Outbreak portal (git)

CNN X-ray disease detection (97%) + NLP fake-news classifier (92%), with COVID tracking dashboard.

LLM Inference Profiling

Roofline decode-latency + KV concurrency model (3B outservices 4B 5x) - vLLM, torch.profiler, AWQ/FP8, CUDA graphs.

Small-Model Specialisation

QLoRA fine-tune with a three-tier evaluation harness - PEFT, TRL, LLM-as-judge, Cohen's kappa, McNemar, eval CI gate.

EDUCATION

COIMBATORE INSTITUTE OF TECHNOLOGY

M.Sc. Decision and Computing Sciences (5 year Integrated) – 9.45 CGPA

Sept 2020 – Apr 2025

Coimbatore, India

PUBLICATION:

Logistics Optimization for E-commerce using Predictive Analytics and Micro Warehousing:

International Journal of Progressive Research in Engineering Management and Science (IJPREMS).

- Reduced delivery overhead by 20% in e-commerce logistics through demand forecasting and spatial data clustering.

EXTRA-CURRICULAR ACTIVITIES AND ACHIEVEMENTS

- Conducted Big Data sessions using PySpark for 60+ students at Vellammal Engineering College, mentoring them in distributed processing & analytics.
- Tata Virtual Internship:** Performed **EDA** on 100,000+ data points using Matplotlib for data visualization to extract user trends and key insights.
- PitchDeck, Runner-up :** Presented GenAlpha, an AI chatbot concept combining rule-based logic and **AGI** to enable personalized interactions.
- Tell a Tale, Melinia :** Secured 2nd prize in encryption and BI-analytics event, presenting insights and business logic using data.
- Solved **450+** problems in **Leetcode**, Earned **3 Gold badges** and **3 certifications** in **Hackerrank** ([Link](#)).